

BRAC University
CSE 350 (Summer 2025)
Section 17
Quiz-2

Name :

ID :

Problem 1:

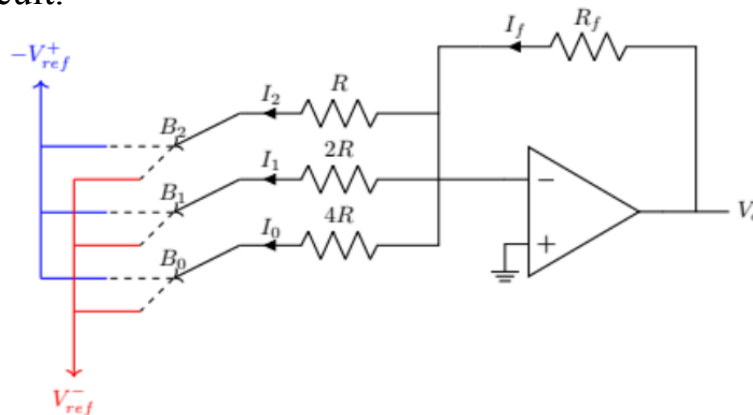
(10 Marks)

Draw a 4-bit **R2R**-DAC (2nd variant). Use **Vref** as **12** volts.

Let **Rf = R**,

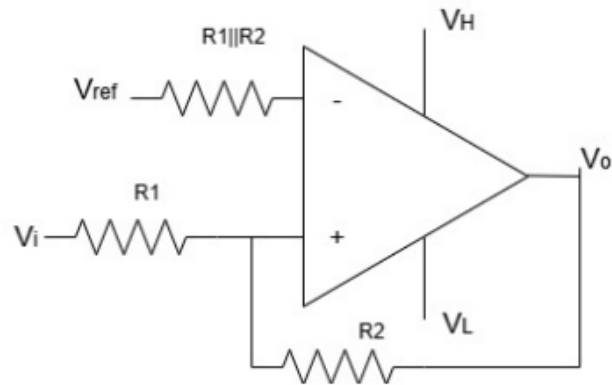
(a) If the input is **0101** what will be the analog output of that circuit?

(b) If you want to design a DAC that can give both negative and positive values as outputs, which type of DAC will you use? For this circuit below, if the LSB output voltage is **-4.5** and the MSB output voltage is **+6.5** then what is the value of reference voltages of the circuit.



Problem 2:

(10 Marks)



$R_1 = 10 \text{ k}\Omega$, $R_2 = 20 \text{ k}\Omega$, $V_{ref} = 4 \text{ V}$, $V_H = 15 \text{ V}$, $V_L = -8 \text{ V}$

Draw the **VTC** curves for this circuit.

- First without the reference voltage and then
- Then with the reference voltage. Does the hysteresis width change?

